Alkane

121. Which of the following is not linked with methane

(a) Marsh gas

(b) Natural gas

(c) Producer gas

(d) Coal gas

122. Which of the following has highest octane number

(a) *n*-hexane

(b) *n*-heptane

(c) *Iso*-octane

(d) *n*-heptane and iso-octane mixed in ratio 50 : 50

123. A mixture of ethyl iodide and *n*-propyl iodide is subjected to Wurtz reaction. The hydrocarbon that will not be formed is

(a) *n*-butane (b) *n*-propane

(c) *n*-pentane (d) *n*-hexane

124. Most of the hydrocarbons from petroleum are obtained by

(a) Fractional distillation

(b) Fractional crystallization

(c) Vaporization

(d) Polymerization

125. Which is the best antiknock compound or Which one of the following substances is used as an antiknock compound

(a) Lead tetrachloride

(b) Lead acetate

(c) Zinc ethyl

(d) Tetraethyl lead (TEL)

126. In the dichlorination reaction of propane, mixture of products are obtained. How many isomers, the mixture contains

(a) 2 (b) 3

(c) 4 (d) 5

127. Which of the following cycloalkane gives open chain compound, when reacts with bromine

(a) Cyclopropane

(b) Cyclopentane

(c) Cyclohexane

(d) Cyclo-octane

128. Grignard reagent is not prepared in aqueous medium but prepared; in ether medium because the reagent

(a) Reacts with water

(b) Is insoluble in water

(c) Is highly reactive in ether

(d) Becomes inactive in water

129. A sample of petrol is a mixture of 30% *n*-heptane and 70% *iso*-octane. The sample has octane number

(a) 30 (b) 70

(c) 15 (d) 35

130. For the reduction of ketones to hydrocarbon, the appropriate agent is

(a) HI

(b) $Zn - Hg/HCl$

(c) Red phosphorous

(d) H_2SO_4

131. Heating of alkanes with fuming sulphuric acid or oleum at high temperature, which forms sulphonic acid, is called

(a) Nitration

(b) Halogenation

(c) Sulphonation

(d) Oxidation

132. Propane is obtained from propene by which method ?

(a) Catalyst hydrogenation

(b) Wurtz reaction

(c) Dehydrogenation

(d) Frankland reaction

133. B.P. of branched chain alkanes as compared to straight chain alkanes are

(a) Lower

(b) Equal

(c) Higher

(d) Independent of the chain

134. Daily use candles (paraffin wax) contain

(a) Higher saturated hydrocarbon

(b) Lower saturated hydrocarbon

(c) Higher unsaturated hydrocarbon

(d) Lower unsaturated hydrocarbon

135. The reaction $CH_4 + Cl_2 \xrightarrow{\text{uv light}} CH_3Cl + HCl$ is an example of

(a) Addition reactions

(b) Substitution reaction

(c) Elimination reaction

(d) Rearrangement reaction

136. Normal butane convert into isobutane by

(a) $LiAlH_4$ (b) $AlCl_3$

(c) $NaBH_4$ (d) Zn/HCl

137. Alcoholic solution of *KOH* is used for
(a) Dehydration
(b) Dehydrogenation
(c)Dehydrohalogenation
(d) Dehalogenation

138. Aluminium carbide on reacting with water gives
(a) Methane (b) Ethane
(c) Ethene (d) Ethyne

139. Maximum carbon-carbon bond distance is found in
(a) Ethyne (b) Ethene
(c) Ethane (d) Benzene

140. Which of the following reaction is expected to readily give a hydrocarbon product in good yields
(a) $RCOOK \xrightarrow{\text{Oxidation Electrolysis}}$
(b) $RCOOAg \xrightarrow{I_2}$
(c) $CH_3 - CH_3 \xrightarrow[hv]{Cl_2}$
(d) $(CH_3)_2CCl \xrightarrow{C_2H_5OH}$

141. Out of the following fractions of petroleum, the one having the lowest boiling point is or Which of the following is obtained at lowest temperature by fractional distillation of petroleum
(a) Kerosene (b) Diesel oil
(c) Gasoline (d) Heavy oil

NEET IIT-JEE BOARD G10 EDUCATIONAL PLATFORM ESTD: 2005